

Memory Maze - Human Vs Ai Pathfinding Game Design Document

Version 1.0



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Revision History

Date	Version	Description	Author
12 / 01 / 2026	1.0	Introduction	BC220407371

Viewer's Guide

It is advised to view this document in “Dark Mode” as the light colored diagrams won’t be visible on “Light Mode”.

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Introduction to Design Document

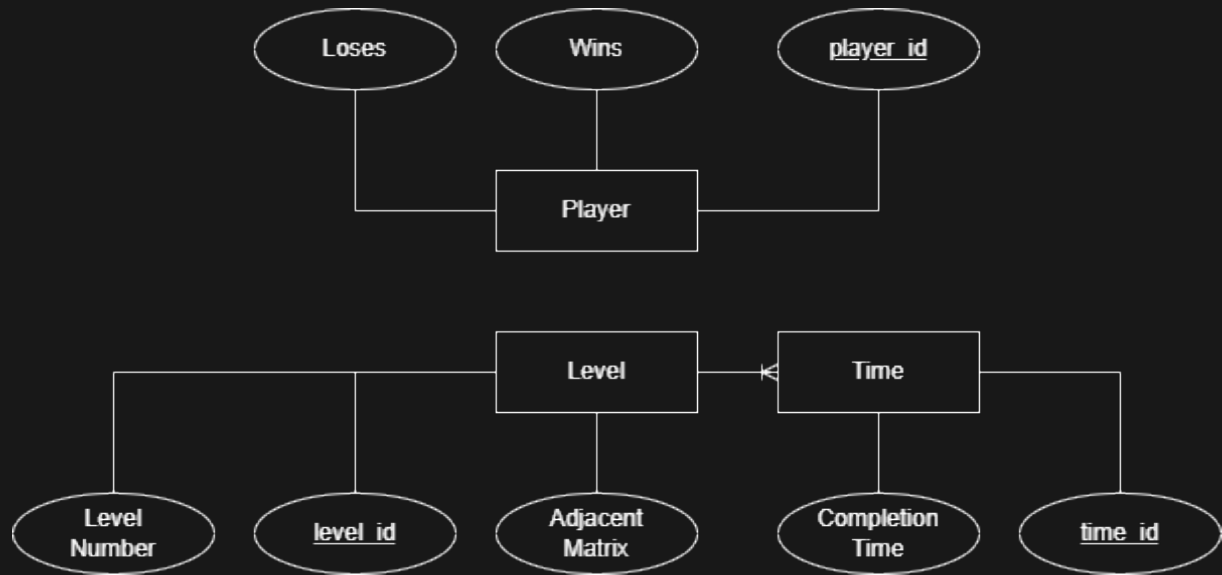
This Design Document describes the detailed system design of “Memory Maze – Human vs AI Pathfinding Game”, developed as a Final Year Project. It explains how the requirements defined in the Software Requirements Specification (SRS) are converted into a structured and implementable system design.

This document includes the

- Entity Relationship Diagram (ERD) to define the data structure
- Sequence diagrams to illustrate interactions among system components
- An architecture design diagram to present the overall system organization
- Class diagrams to represent the object-oriented structure of the system.
- Database design for data storage
- Interface design to describe user interaction
- Test cases to validate the correctness of the proposed design

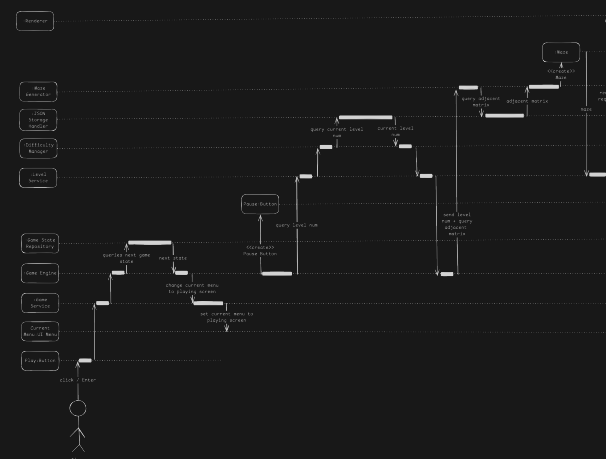
The purpose of the design phase is to establish a clear blueprint for development by identifying system components, defining their responsibilities, and specifying interactions before

By providing visual and structural representations of the system, the design phase enhances communication among developers and evaluators and ensures efficient and reliable implementation of “Memory Maze – Human vs AI Pathfinding Game”.

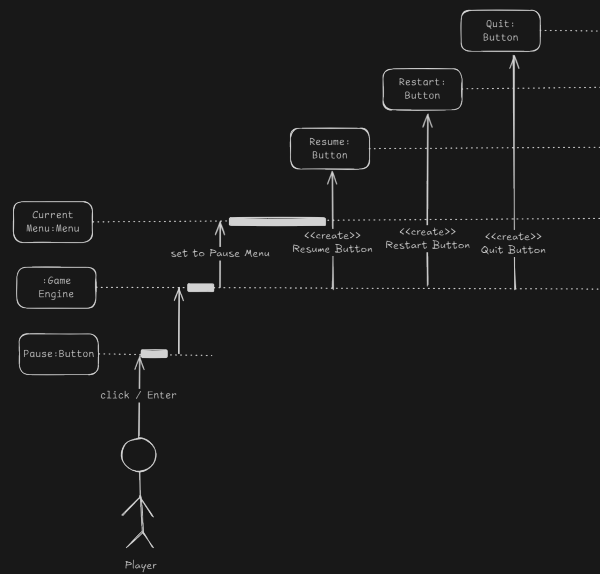


Provide Sequence Diagrams for

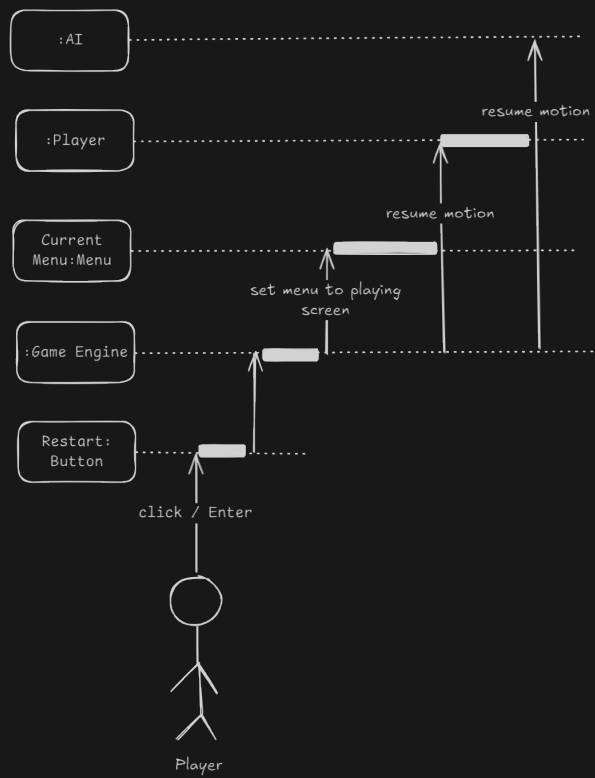
Provide Sequence Diagrams for each of the use case to show the task sequence of the system,
Provide a sequence diagram for each of the use case provided in use case diagram



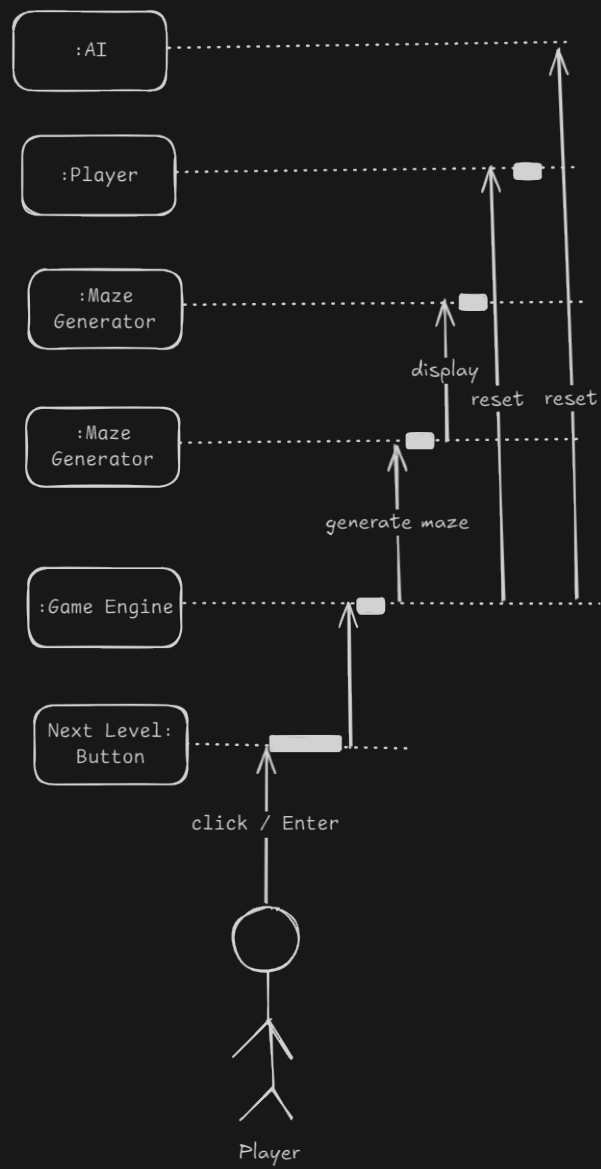
Pause Game



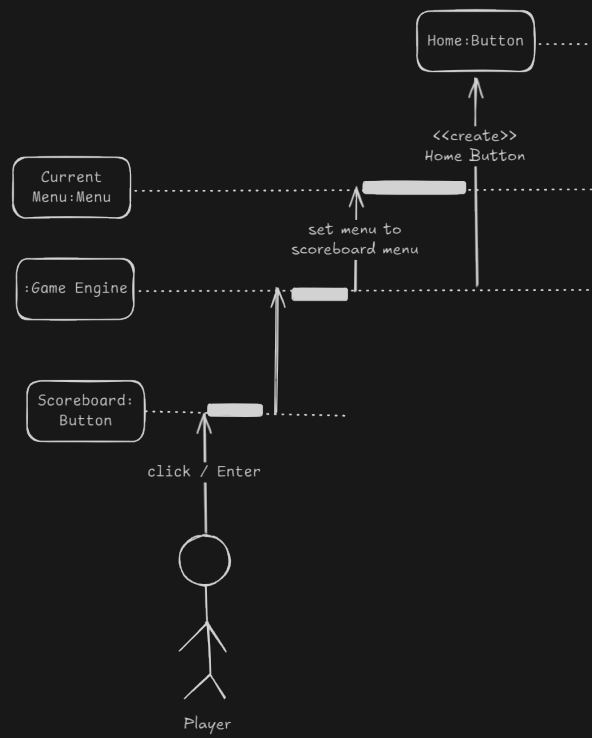
Restart Game



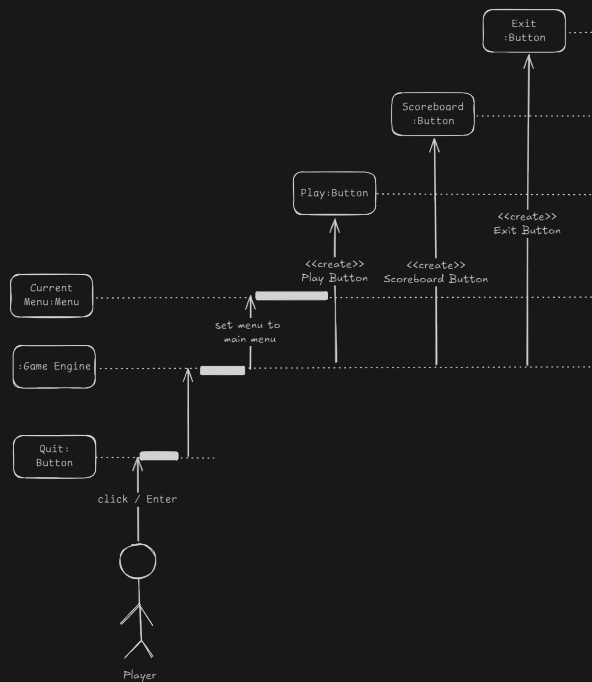
Advance to next Level



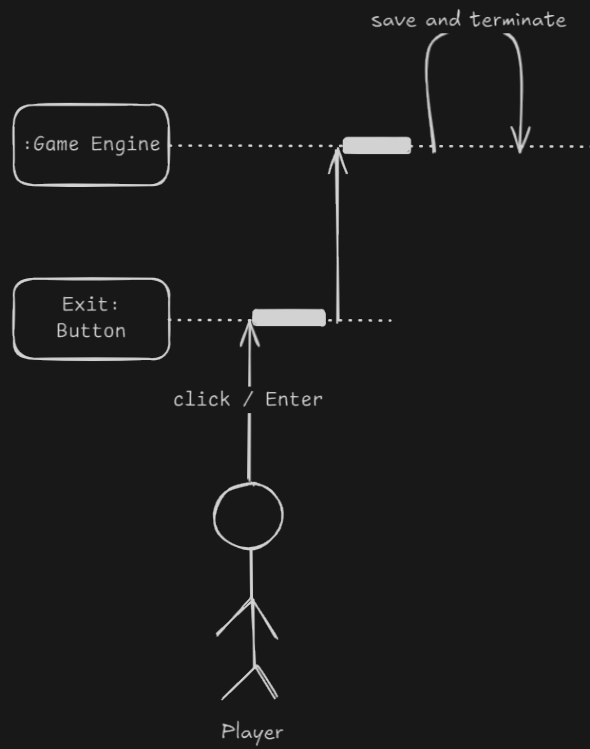
View Scores



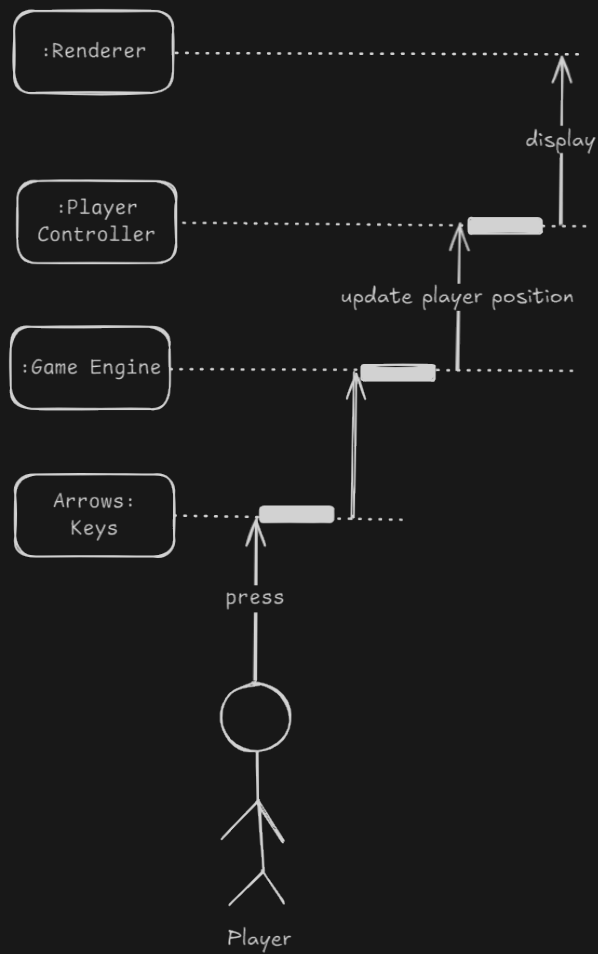
Quit



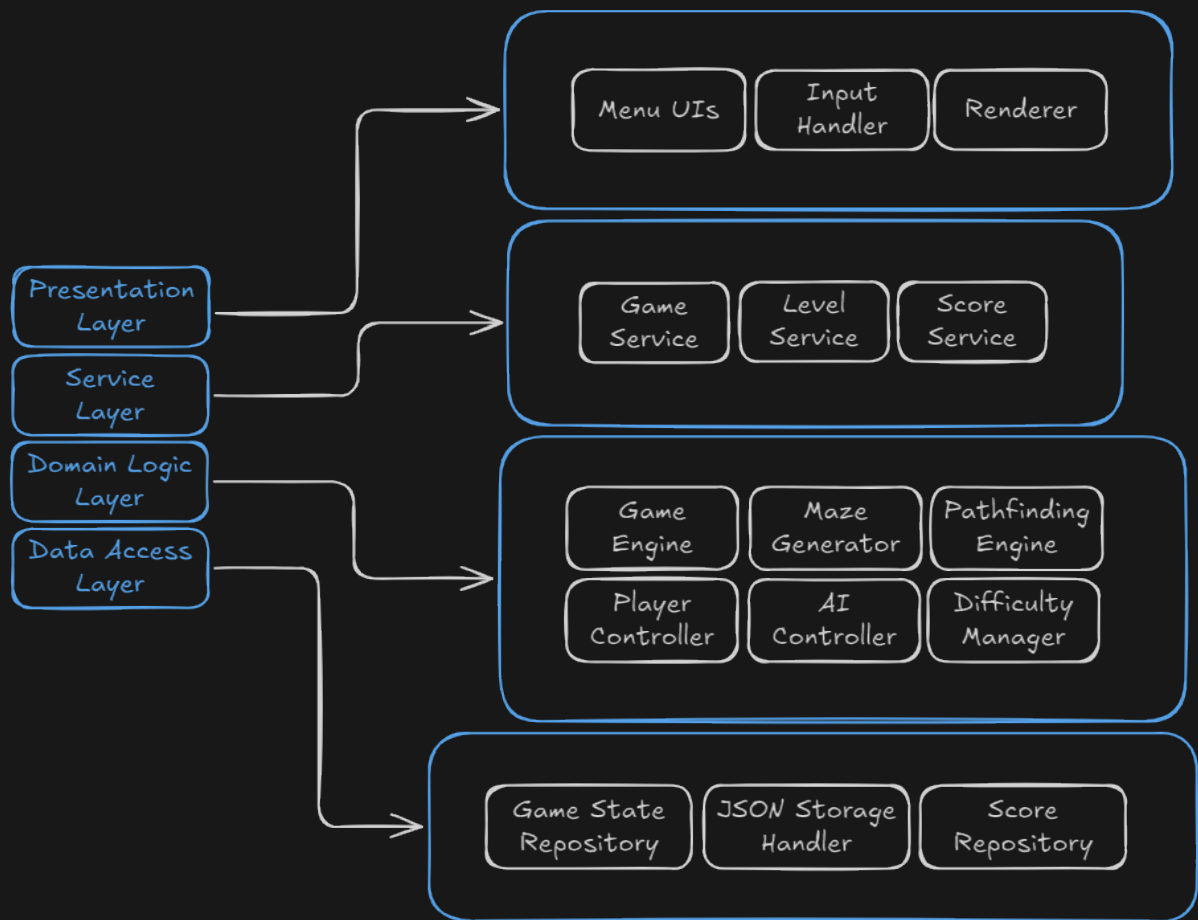
Exit



Navigate Maze



Architecture Design Diagram



Class Diagram



Database Design

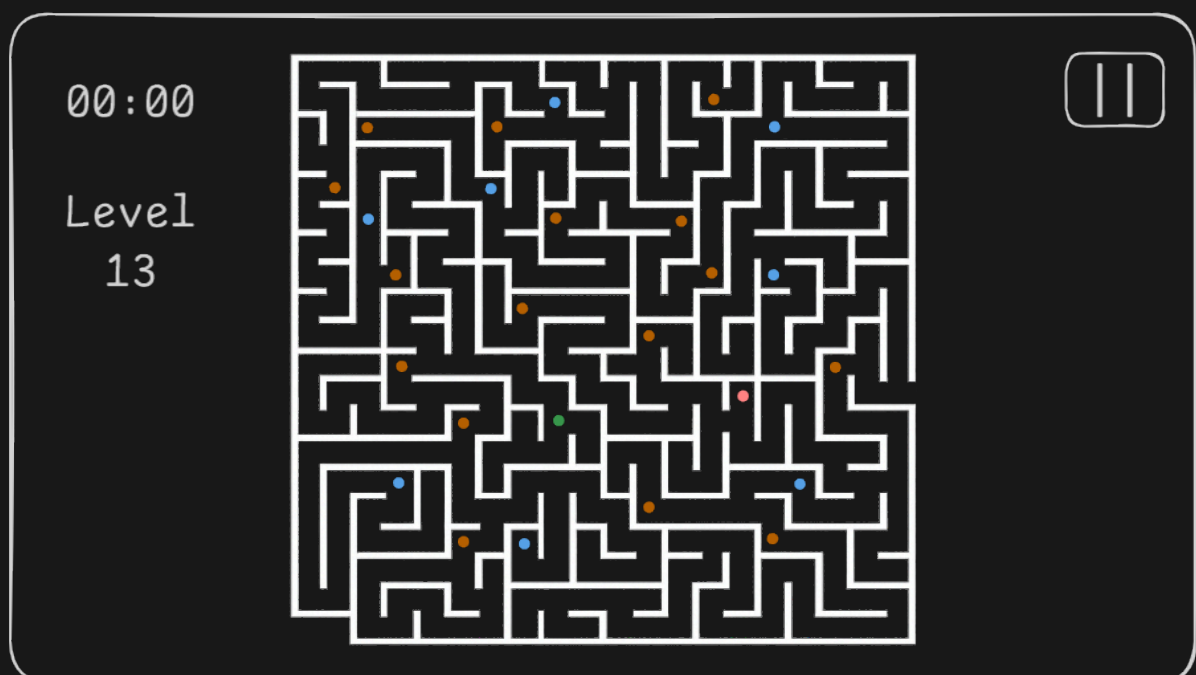
“Memory Maze - Human vs AI Pathfinding Game” doesn’t use a database but rather a JSON file as a datastore.

Interface Design

Main Menu



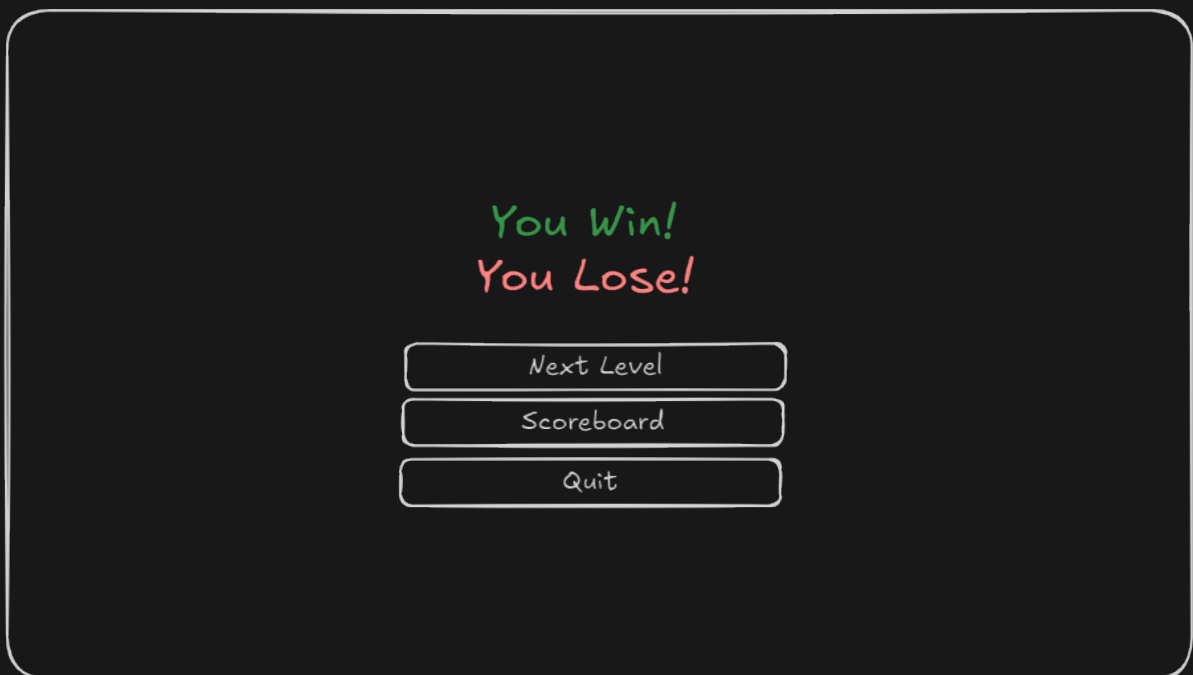
Playing



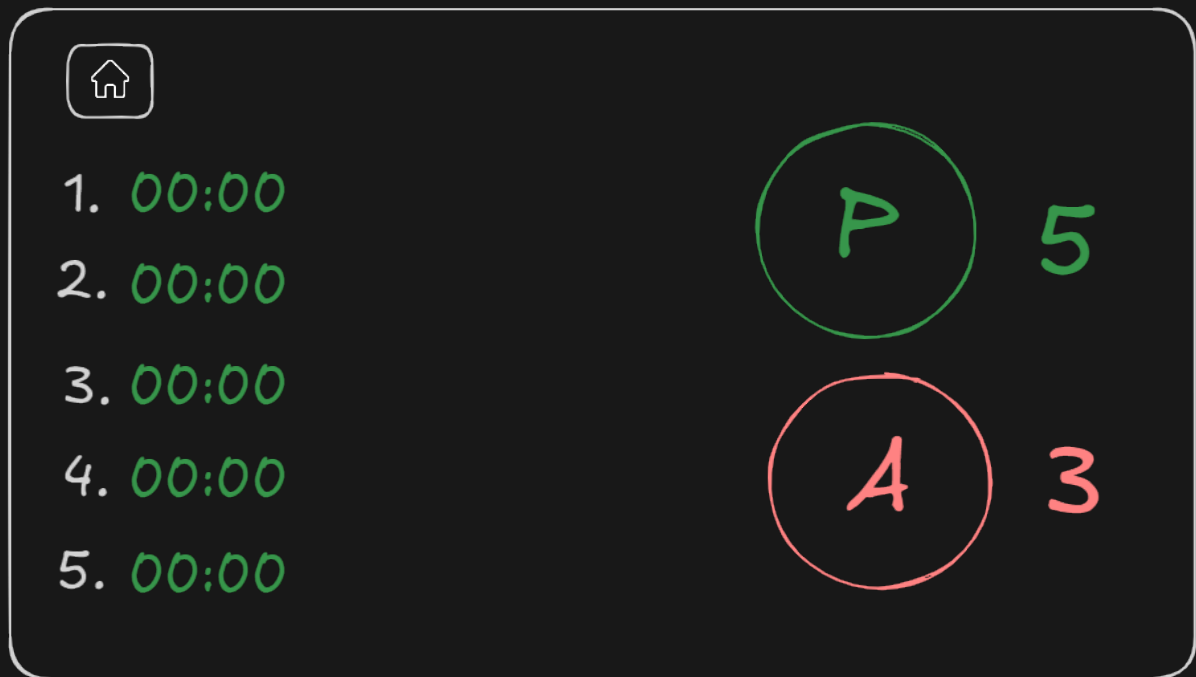
Pause Menu



Result Menu



Scoreboard



Test Cases

Test case 1

Test Case Title: Testing Play button which starts the game.

Fields	Values
Preconditions	Current menu: Main menu
Actions	Enter key press or left mouse click on Play button.
Expected Results	Game starts.
Tested By	Muhammad Zaid
Result	

Test case 2

Test Case Title: Testing Scoreboard button which shows the wins and losses.

Fields	Values
Preconditions	Current Menu: Main menu or Result menu
Actions	Enter key press or left mouse click on Scoreboard button.
Expected Results	Take user to Scoreboard menu
Tested By	Muhammad Zaid
Result	

Test case 3

Test Case Title: Testing exit button which saves and terminates the game.

Fields	Values
Preconditions	Current menu: Main menu
Actions	Enter key press or left mouse click on Exit button.
Expected Results	Save data and terminate the game
Tested By	Muhammad Zaid
Result	

Test case 4

Test Case Title: Testing navigation and making sure player moves in the directions corresponding to the arrow keys on the keyboard.

Fields	Values
Preconditions	Current Game State: Playing and there is no wall in front of player.
Actions	Press arrow keys
Expected Results	Player should move in the corresponding direction
Tested By	Muhammad Zaid
Result	

Test case 5

Test Case Title: Testing Pause button which pauses the game while it is being played.

Fields	Values
Preconditions	Current Game State: Playing
Actions	Esc key press or left mouse click on Pause button.
Expected Results	Take user to Pause menu
Tested By	Muhammad Zaid
Result	

Test case 6

Test Case Title: Testing Resume button which resumes the game without changing anything.

Fields	Values
Preconditions	Current menu: Pause menu
Actions	Enter key press or left mouse click on Resume button.
Expected Results	Take user to back to playing state
Tested By	Muhammad Zaid
Result	

Test case 7

Test Case Title: Testing Restart button which resets the level being currently played.

Fields	Values
Preconditions	Current menu: Pause menu
Actions	Enter key press or left mouse click on Restart button.
Expected Results	Reset everything on playing screen except for maze structure
Tested By	Muhammad Zaid
Result	

Test case 8

Test Case Title: Testing Next Level button which advances to the next level.

Fields	Values
Preconditions	Current Menu: Result Menu
Actions	Enter key press or left mouse click on Next Level button.
Expected Results	Advances user to next level
Tested By	Muhammad Zaid
Result	

Test case 9

Test Case Title: Testing quit button which takes to the main menu.

Fields	Values
Preconditions	Current menu: Pause menu or Result menu
Actions	Enter key press or left mouse click on Quit button.
Expected Results	Take user to main menu
Tested By	Muhammad Zaid
Result	